Cairo University

Faculty of Engineering

Electronics and Electrical Communications Engineering

Assignment 1

ELC 4028 - Artificial Neural Networks

ID	*Sec*	*Name*
9220399	**2**	**شهد جمال محمود حسين**
9220411	**2**	**صلاح الدين حسن حسن محمد**
9220984	**4**	**يوسف خالد عمر محمود**

Supervised by:

Dr. Mohsen Rashwan

Dr. Mohamed Motaz

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PART1 : Regressing Problems

Problem 1: Insurance Company

The numbers of insured persons y with an insurance company for the years 1987 to 1996 are given. We let x represent the number of years since 1987 (x = 0 for 1987, x = 9 for 1996).

1.1 Dataset

x (years since 1987)	Year	y (insured persons)
0	1987	12,400
1	1988	10,900
2	1989	10,000
3	1990	**1,050 (OUTLIER)**
4	1991	9,500
5	1992	8,900
6	1993	8,000
7	1994	7,800
8	1995	7,600
9	1996	7,200

** Value 1050 at x = 3 (year 1990) is the outlier — likely a data entry error or anomaly.*

1.2 Case A: WITH Outlier

1. Scatter Plot & Outlier Identification

A scatter plot of y (number of insured persons) versus x (years since 1987) reveals a generally decreasing trend from ~12,400 down to ~7,200. However, one data point stands out sharply:

- x = 3 (year 1990): y = 1050

This value is dramatically lower than the surrounding data , which are all within the range of approximately **7,200 to 12,400**. The sharp drop at this point breaks the otherwise smooth downward trend.

Therefore, this observation is identified as a clear outlier, most likely due to a data entry error (for example, a missing digit such as *10,500* instead of *1,050*).

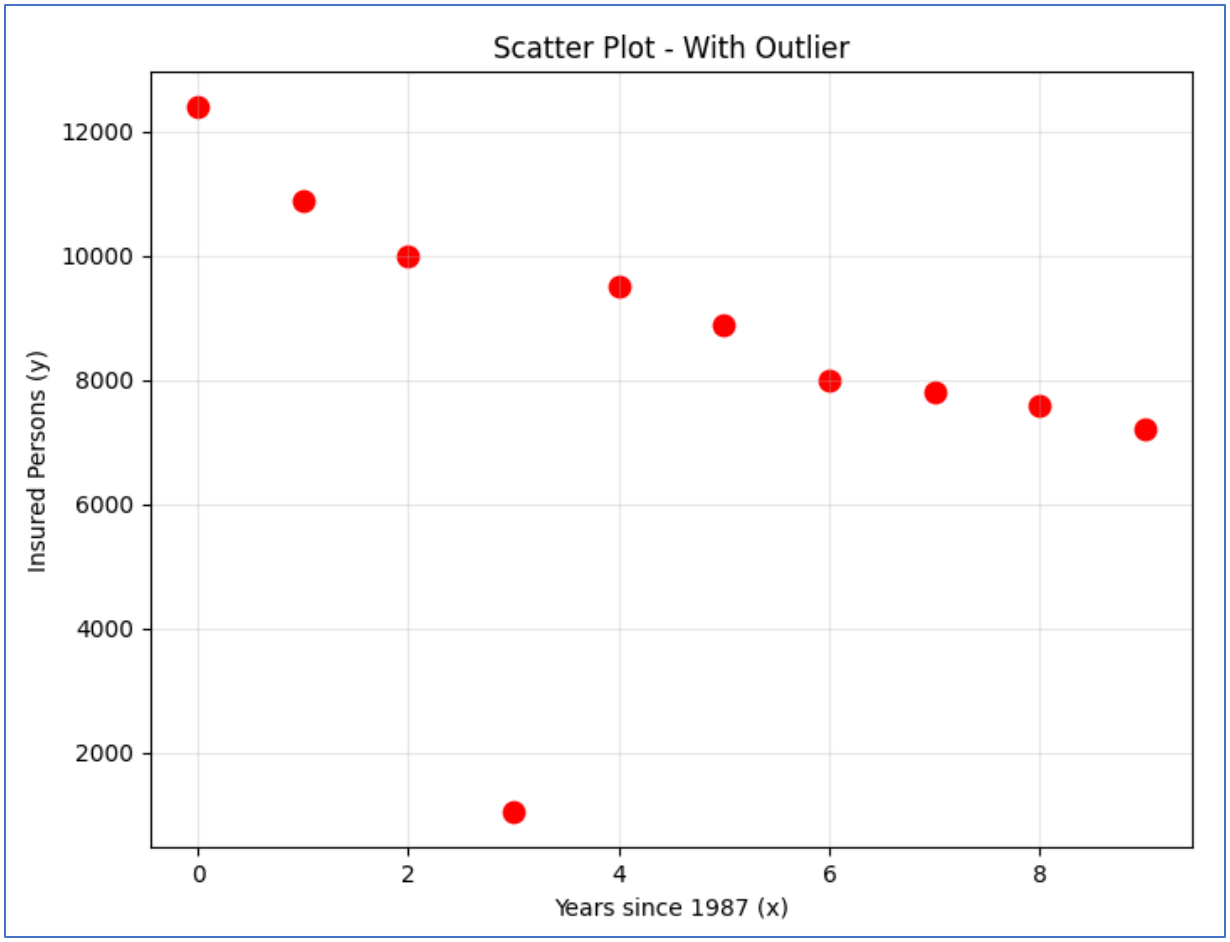

Figure 1: Scatter plot of original dataset identifying the outlier at x = 3 (y = 1050)

2. Regression Models (WITH Outlier)

We apply both OLS and matrix-form methods to compute linear model parameters, then use the matrix form for quadratic and cubic models.

2.1 Linear Model

➢ Hand Analysis (OLS Method)

The equation for the linear model is given as follows:

$$\widehat{y_i} = \beta_0 + \beta_1 x_i$$

The key formulas used are:

$$\widehat{\beta_1} = \frac{\sum_{i=i}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}} \quad , \quad \widehat{\beta_0} = \hat{y} - \widehat{\beta_i} x$$

Data Table for Hand Computation:

year	x_i	y_i	x_i^2	y_i^2	x_iy_i
1987	0	12400	0	153,760,000	0
1988	1	10900	1	118,810,000	10,900
1989	2	10000	4	100,000,000	20,000
1990	3	1050	9	1,102,500	3,150
1991	4	9500	16	90,250,000	38,000
1992	5	8900	25	79,210,000	44,500
1993	6	8000	36	64,000,000	48,000
1994	7	7800	49	60,840,000	54,600
1995	8	7600	64	57,760,000	60,800
1996	9	7200	81	51,840,000	64,800
Σ =	**45**	**83,350**	**285**	**777,572,500**	**344,750**

$$n = 10 \quad , \quad y = \frac{83350}{10} = 8335 \quad , \quad x = \frac{45}{10} = 4.5$$

$$\widehat{\beta_1} = \frac{344{,}750 - \frac{45 * 83{,}350}{10}}{285 - \frac{45^2}{10}} = -367.57$$

$$\widehat{\beta_0} = 8335 + 367.57 * 4.5 = 9989.09$$

➢ Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \qquad X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}, Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 9989.09 \\ -367.576 \end{pmatrix}$$

So ,linear equation is:

$$\hat{y}_i = 9989.09 - 367.576\, x$$

➢ R² Verification Table

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{83350}{10} = 8335$$

year	x_i	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i$	SSE = $(y_i - \hat{y}_i)^2$	SSyy = $(y_i - \bar{y})^2$
1987	0	12400	9,989.1	5,812,483	16,524,225
1988	1	10900	9,621.5	1,634,524	6,579,225
1989	2	10000	9,253.9	556,606	2,772,225
1990	3	1050	8,886.4	61,408,600	53,071,225
1991	4	9500	8,518.8	962,777	1,357,225
1992	5	8900	8,151.2	560,683	319,225
1993	6	8000	7,783.6	46,813	112,225
1994	7	7800	7,416.1	147,410	286,225
1995	8	7600	7,048.5	304,169	540,225
1996	9	7200	6,680.9	269,455	1,288,225
Σ =	**45**	**83,350**	**83,350**	**71,703,515**	**82,850,250**

$$R^2_{linear\ model} = \frac{SS - SSE}{SS} = \frac{82{,}850{,}250 - 71{,}703{,}515}{82{,}850{,}250} = 0.13454$$

➢ Output Graph

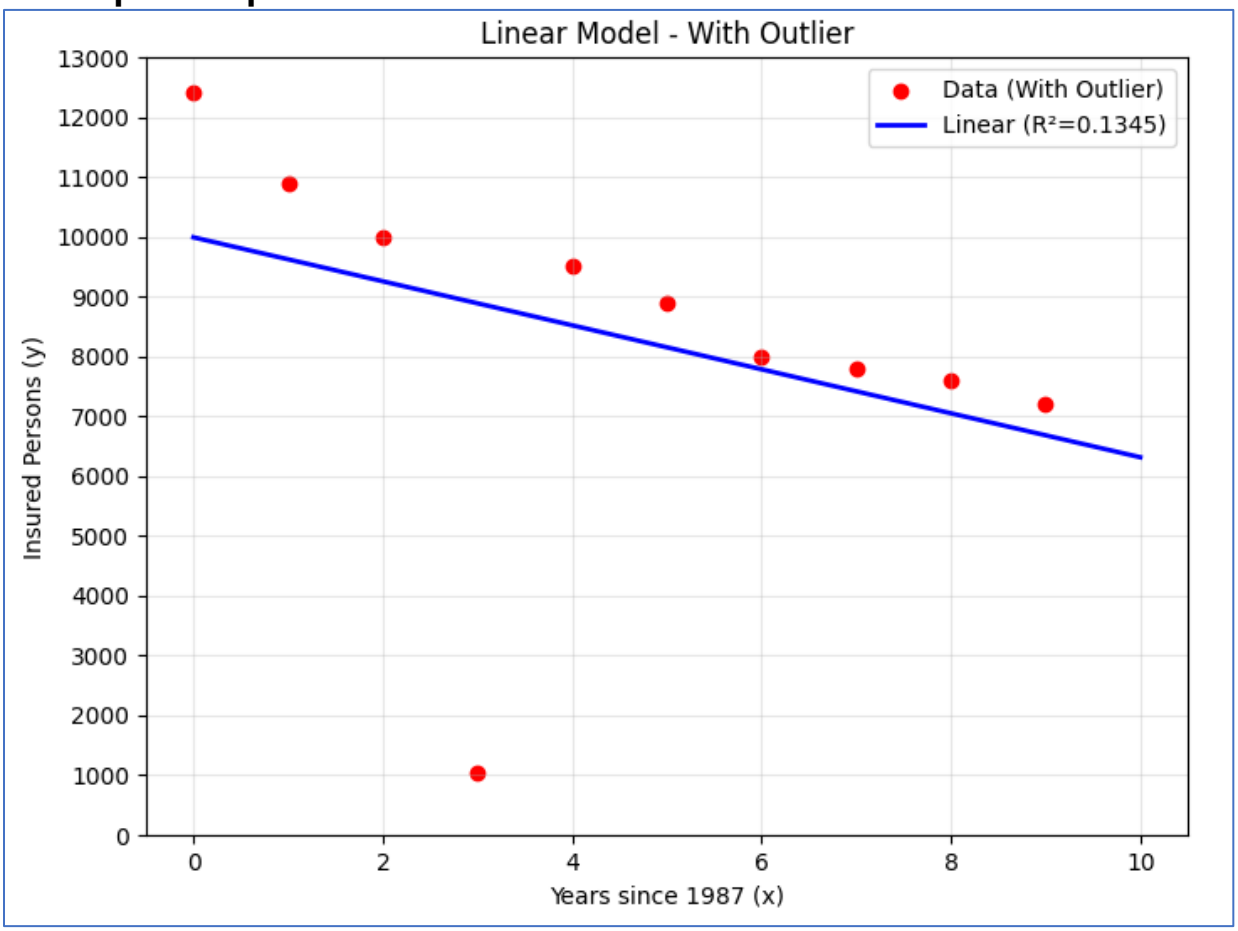

Figure 2: Linear Model of Original Data with Outlier

➢ Simulation Results:

```
Linear Model:
  Equation: y = 9989.09 + (-367.58)*x^1
  R^2: 0.1345
```

Figure 3: Simulation Results of Linear Model of Original Data with Outlier

2.2 Quadratic Model

The equation for the quadratic model is given as follows:

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$$

➢ Hand Analysis (Matrix Form)

$$\beta = (X^T X)^{-1} X^T Y$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}, \qquad \mathrm{X} = \begin{pmatrix} 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \\ 1 & 6 & 6^2 \\ 1 & 7 & 7^2 \\ 1 & 8 & 8^2 \\ 1 & 9 & 9^2 \end{pmatrix}, Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 11627.73 \\ -1596.55 \\ 136.55 \end{pmatrix}$$

So ,linear equation is:

$$\hat{y}_i = 11627.73 - 1596.55x + 136.55x^2$$

➢ R² Verification Table

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{83350}{10} = 8335$$

year	x_i	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$	SSE = $(y_i - \hat{y}_i)^2$	SSyy = $(y_i - \bar{y})^2$
1987	0	12400	**11,627.7**	**596,405**	**16,524,225**
1988	1	10900	10,167.7	536,223	6,579,225
1989	2	10000	8,980.8	1,038,701	2,772,225
1990	3	1050	8,067.0	49,238,925	53,071,225
1991	4	9500	7,426.4	4,299,969	1,357,225

1992	5	8900	7,058.8	3,390,064	319,225
1993	6	8000	6,964.3	1,072,639	112,225
1994	7	7800	7,143.0	431,710	286,225
1995	8	7600	7,594.7	28	540,225
1996	9	7200	8,319.5	1,253,377	1,288,225
Σ =	**45**	**83,350**	83,350	61,858,042	82,850,250

$$R^2_{quad\ model} = \frac{SS - SSE}{SS} = \frac{82{,}850{,}250 - \mathbf{61{,}858{,}042}}{82{,}850{,}250} = 0.2534$$

➢ Output Graph

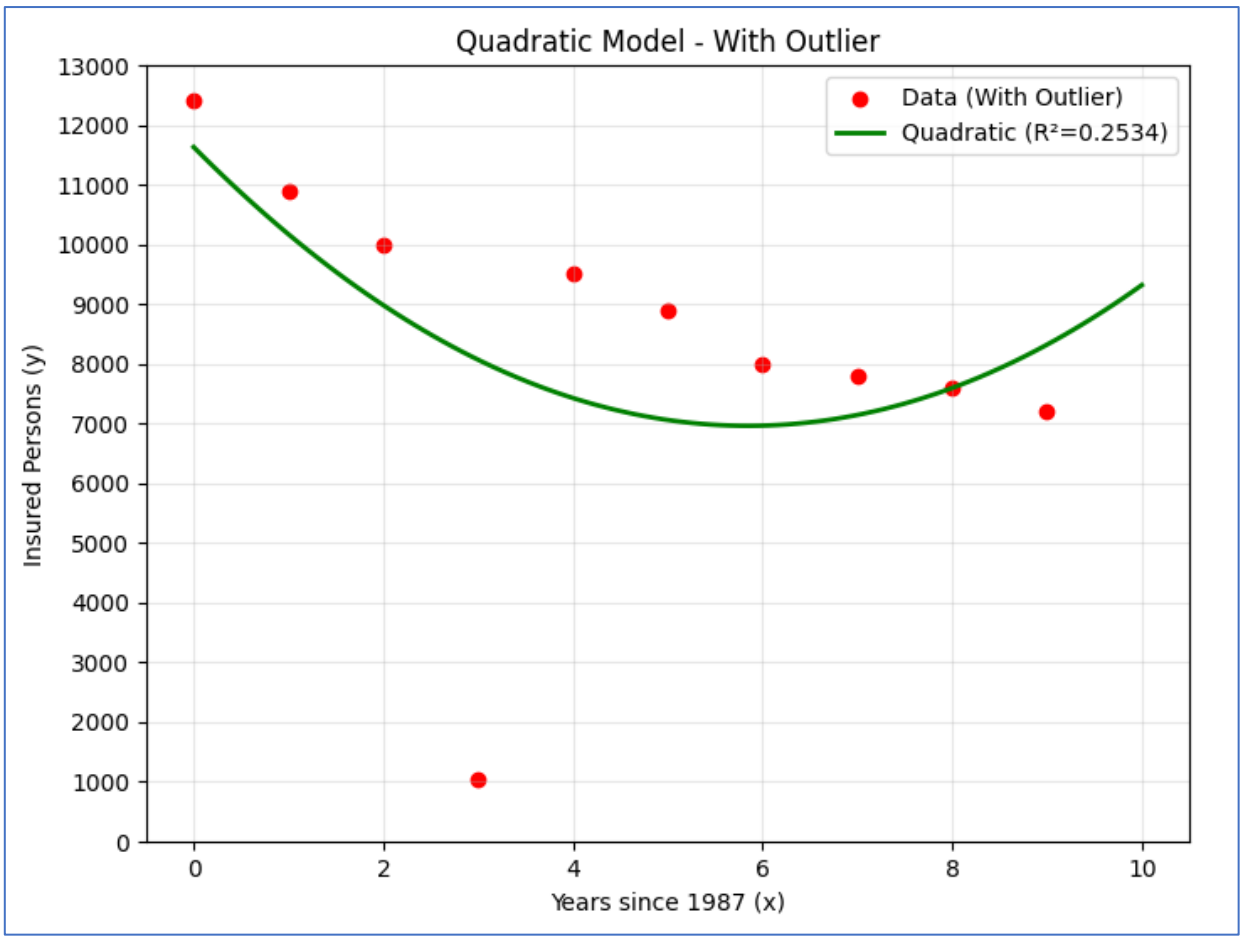

Figure 4: Quadratic Model of Original Data with Outlier

➢ Simulation Results:

```
Quadratic Model:
  Equation: y = 11627.73 + (-1596.55)*x^1 + (136.55)*x^2
  R^2: 0.2534
```

Figure 5: Simulation Results of Quadratic Model of Original Data with Outlier

2.3 Cubic Model

The equation for the cubic model is given as follows:

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$$

➢ Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y}$$

$$\boldsymbol{\beta} = \begin{pmatrix} \boldsymbol{\beta}_0 \\ \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \boldsymbol{\beta}_3 \end{pmatrix}, \quad \mathrm{X} = \begin{pmatrix} 1 & 0^2 & 0^3 \\ 1 & 1^2 & 1^3 \\ 1 & 2^2 & 2^3 \\ 1 & 3^2 & 3^3 \\ 1 & 4^2 & 4^3 \\ 1 & 5^2 & 5^3 \\ 1 & 6^2 & 6^3 \\ 1 & 7^2 & 7^3 \\ 1 & 8^2 & 8^3 \\ 1 & 9^2 & 9^3 \end{pmatrix}, Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \boldsymbol{\beta}_0 \\ \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \boldsymbol{\beta}_3 \end{pmatrix} = \begin{pmatrix} 13113.15 \\ -4313.93 \\ 932.31 \\ -58.95 \end{pmatrix}$$

So ,linear equation is:

$$\hat{y}_i = 13113.15 - 4313.93\, x_i + 932.31\, x_i^2 - 58.95\, x_i^3$$

➢ R² Verification Table

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{83350}{10} = 8335$$

year	x_i	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$	SSE = $(y_i - \hat{y}_i)^2$	SSyy = $(y_i - \bar{y})^2$
1987	0	12400	**13,113.1**	**508,578**	**16,524,225**
1988	1	10900	9,672.6	1,506,542	6,579,225

1989	2	10000	7,743.0	5,094,123	2,772,225
1990	3	1050	6,970.7	35,054,266	53,071,225
1991	4	9500	7,002.0	6,240,214	1,357,225
1992	5	8900	7,483.2	2,007,341	319,225
1993	6	8000	8,060.7	3,684	112,225
1994	7	7800	8,380.8	337,334	286,225
1995	8	7600	8,089.8	239,940	540,225
1996	9	7200	6,834.1	133,864	1,288,225
Σ =	**45**	**83,350**	83,350	51,125,885	82,850,250

$$R^2_{cubic\ model} = \frac{SS - SSE}{SS} = \frac{82{,}850{,}250 - \mathbf{51{,}125{,}885}}{82{,}850{,}250} = 0.3829$$

➢ Output Graph

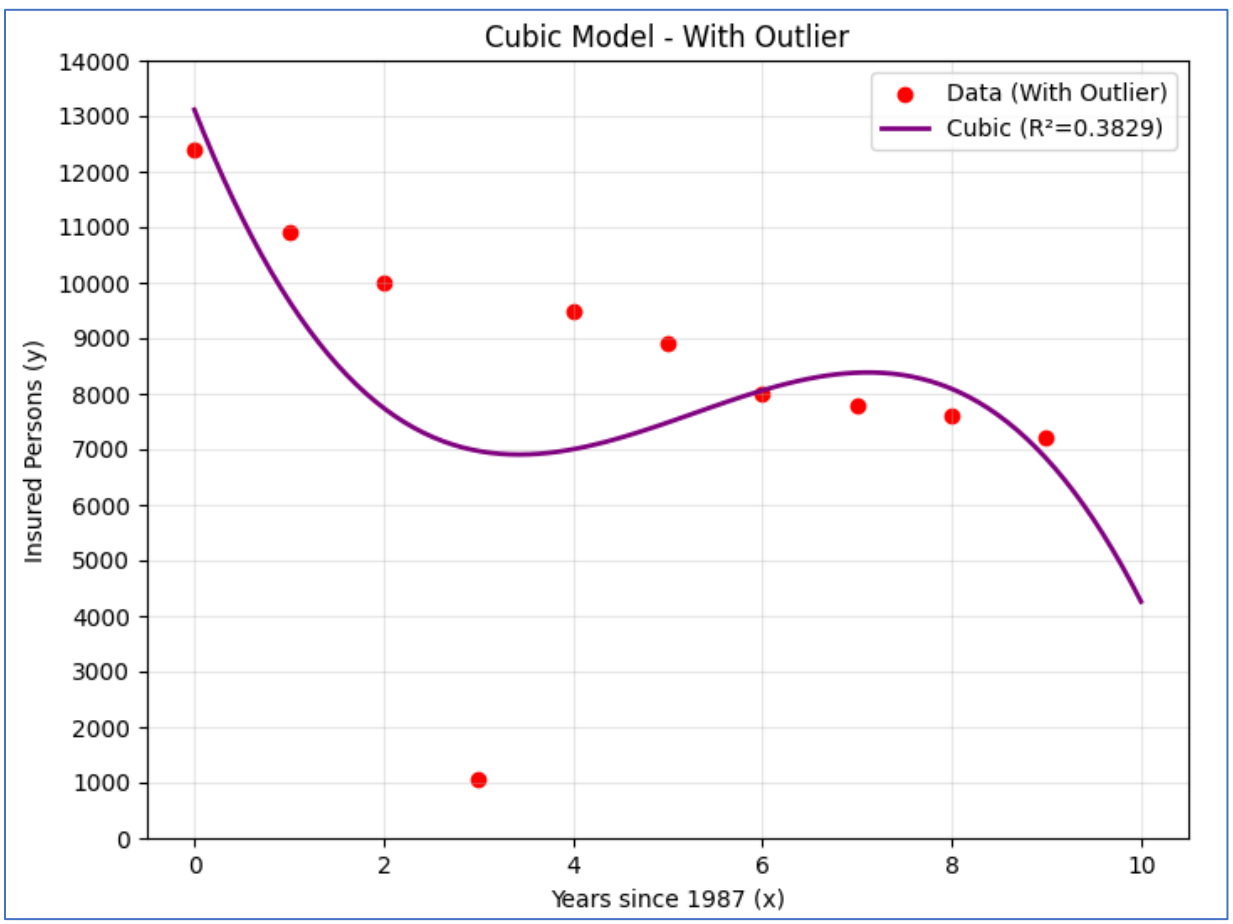

Figure: : Cubic Model of Original Data with Outlier

➢ Simulation Results:

```
Cubic Model:
  Equation: y = 13113.15 + (-4313.93)*x^1 + (932.31)*x^2 + (-58.95)*x^3
  R^2: 0.3829
```

Figure 6: : Simulation Results of Cubic Model of Original Data with Outlier

3. Comparison between three models (WITH Outlier):

Model	Equation (MATLAB)	R^2
Linear	$\hat{y}_i = 9989.09 - 367.576\, x_i$	0.1345
Quadratic	$\hat{y}_i = 11627.73 - 1596.55\, x_i + 136.55\, x_i^2$	0.2534
Cubic	$\hat{y}_i = 13113.15 - 4313.93\, x_i + 932.31\, x_i^2 - 58.95\, x_i^3$	0.3829

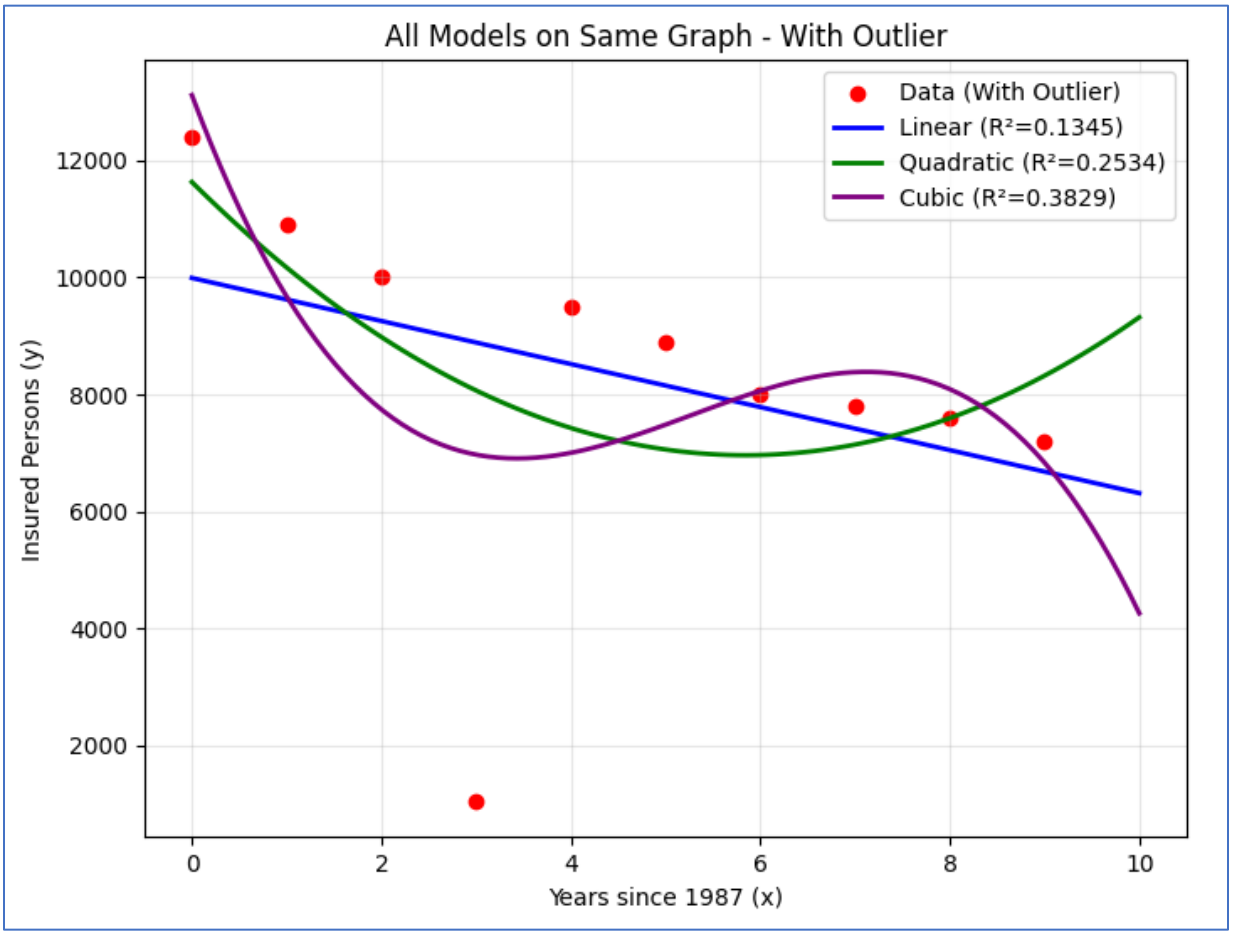

Figure 7: Comparison between three Models of Original Data with Outlier

Comment:

All R^2 values are low (< 0.50) due to the heavy influence of the outlier at x = 3 (y = 1050). The cubic model yields the highest R.

1.3 Case B: WITHOUT Outlier

1. Outlier Removal and Scatter Plot of Cleaned Data

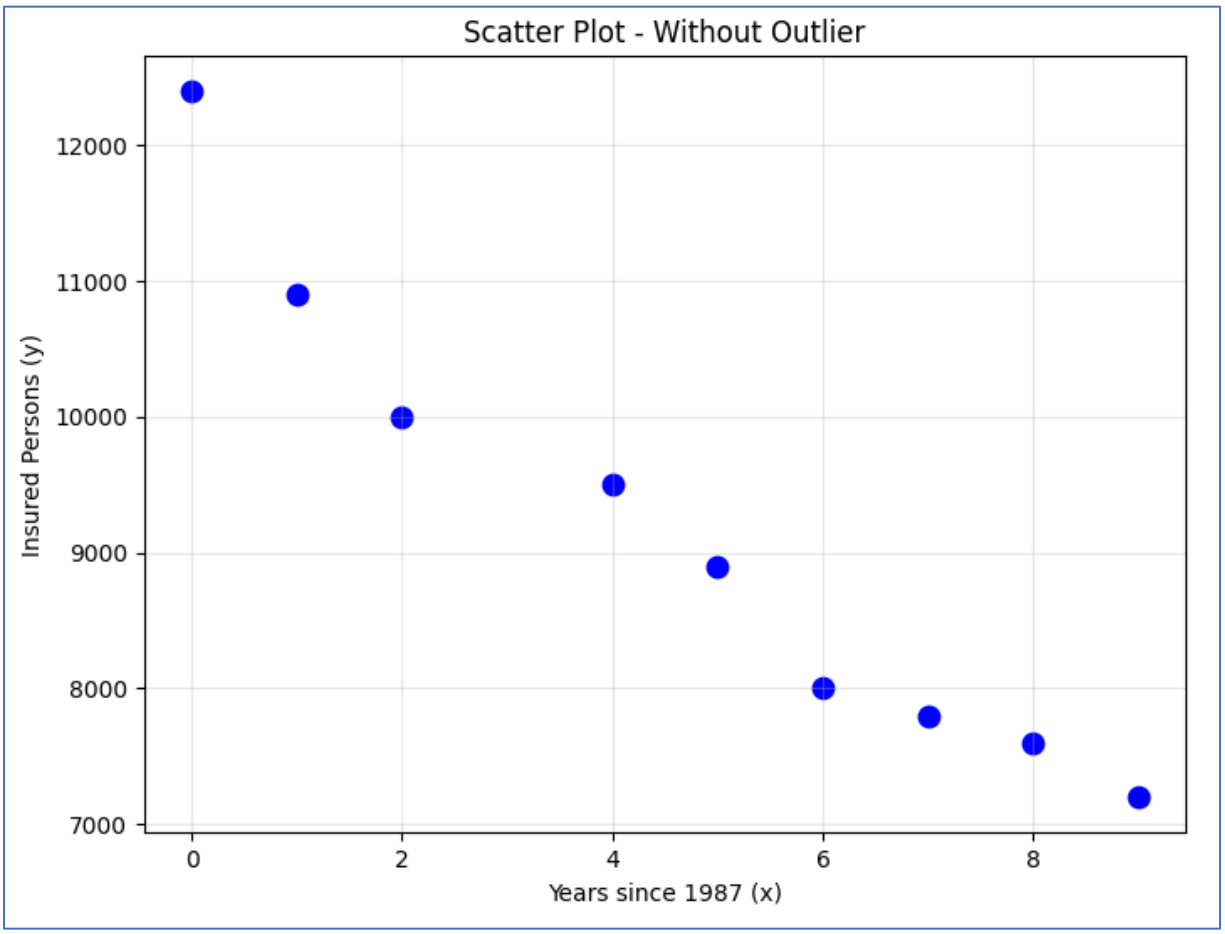

Figure 8: Scatter Plot of Cleaned Data without Outlier

2. Regression Models (WITHOUT Outlier)

After removing the outlier at x = 3 (y = 1,050), the cleaned dataset contains 9 data points. We now repeat the regression analysis using the same methods.

2.1 Linear Model

➢ Hand Analysis (OLS Method)

The equation for the linear model is given as follows:

$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

Data Table for Hand Computation:

year	x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1987	0	12,400	0	153,760,000	0

1988	1	10,900	1	118,810,000	10,900
1989	2	10,000	4	100,000,000	20,000
1991	4	9,500	16	90,250,000	38,000
1992	5	8,900	25	79,210,000	44,500
1993	6	8,000	36	64,000,000	48,000
1994	7	7,800	49	60,840,000	54,600
1995	8	7,600	64	57,760,000	60,800
1996	9	7,200	81	51,840,000	64,800
Σ =	**42**	**82,300**	**276**	**776,470,000**	**341,600**

$$n = 9 \quad , \quad y = \frac{82300}{9} = 9{,}144.44 \quad , \quad x = \frac{42}{9} = 4.667$$

$$\widehat{\beta_1} = \frac{341{,}600 - \frac{42*82300}{9}}{276 - \frac{42^2}{9}} = -530.83$$

$$\widehat{\beta_0} = 9{,}144.44 + 530.83 * 4.667 = 11{,}621.67$$

➢ Hand Analysis (Matrix Form)

$$\beta = (X^T X)^{-1} X^T Y$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}, Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 11{,}621.67 \\ -530.83 \end{pmatrix}$$

So ,linear equation is:

$$\widehat{y_i} = 11{,}621.67 - 530.83\, x$$

➢ R² Verification Table

$$\widehat{y_i} = \frac{82300}{\mathbf{9}} = 9{,}144.44$$

year	x_i	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i$	SSE = $(y_i - \hat{y}_i)^2$	SSyy = $(y_i - \bar{y})^2$
1987	0	12400	11621.7	605,803	10,598,642
1988	1	10900	11090.8	36,417	3,081,975
1989	2	10000	10560.0	313,600	731,975
1991	4	9500	9498.3	3	126,420
1992	5	8900	8967.5	4,556	59,753
1993	6	8000	8436.7	190,678	1,309,753
1994	7	7800	7905.8	11,201	1,807,531
1995	8	7600	7375.0	50,625	2,385,309
1996	9	7200	6844.2	126,617	3,780,864
Σ =	**42**	**82,300**	**82300.0**	**1,339,500**	**23,882,222**

$$R^2_{linear\ model} = \frac{SS - SSE}{SS} = \frac{23{,}882{,}222 - 1{,}339{,}500}{23{,}882{,}222} = 0.9439$$

➢ Output Graph

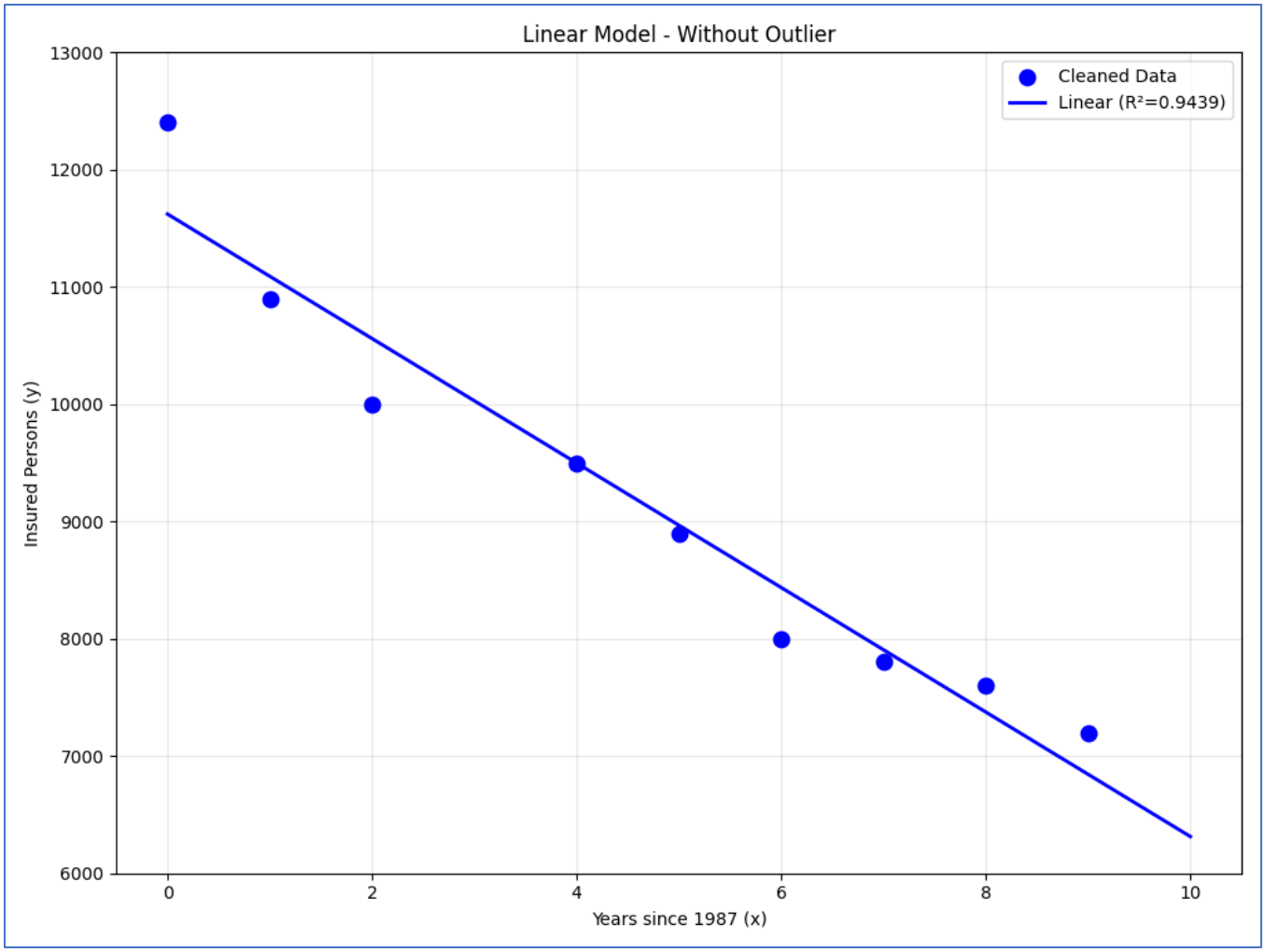

Figure 9: Linear Model of Cleaned Data without Outlier

➢ Simulation Results:

```
Linear Model:
  Equation: y = 11621.67 + (-530.83)*x^1
  R^2: 0.9439
```

Figure 10: Simulation Results of Linear Model of Cleaned Data without Outlier

2.2 Quadratic Model

The equation for the quadratic model is given as follows:

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$$

➢ Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y}$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \\ 1 & 6 & 6^2 \\ 1 & 7 & 7^2 \\ 1 & 8 & 8^2 \\ 1 & 9 & 9^2 \end{pmatrix}, \boldsymbol{Y} = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 12,024.17 \\ -863.13 \\ 37.44 \end{pmatrix}$$

So ,linear equation is:

$$\hat{y}_i = 12,024.17 - 863.13\,x + 37.44\,x^2$$

➢ R² Verification Table

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{82,300}{9} = 9,144.44$$

year	x_i	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$	SSE = $(y_i - \hat{y}_i)^2$	SSyy = $(y_i - \bar{y})^2$
1987	0	12400	12024.2	141,248	10,598,642
1988	1	10900	11198.5	89,090	3,081,975
1989	2	10000	10447.7	200,411	731,975
1991	4	9500	9170.7	108,429	126,420
1992	5	8900	8644.6	65,249	59,753
1993	6	8000	8193.3	37,362	1,309,753
1994	7	7800	7816.9	286	1,807,531
1995	8	7600	7515.4	7,156	2,385,309
1996	9	7200	7288.8	7,884	3,780,864
Σ =	**42**	**82,300**	**82300.0**	**657,116**	**23,882,222**

$$R^2_{quad\ model} = \frac{SS - SSE}{SS} = \frac{23,882,222 - 657,116}{23,882,222} = 0.9725$$

➢ Output Graph

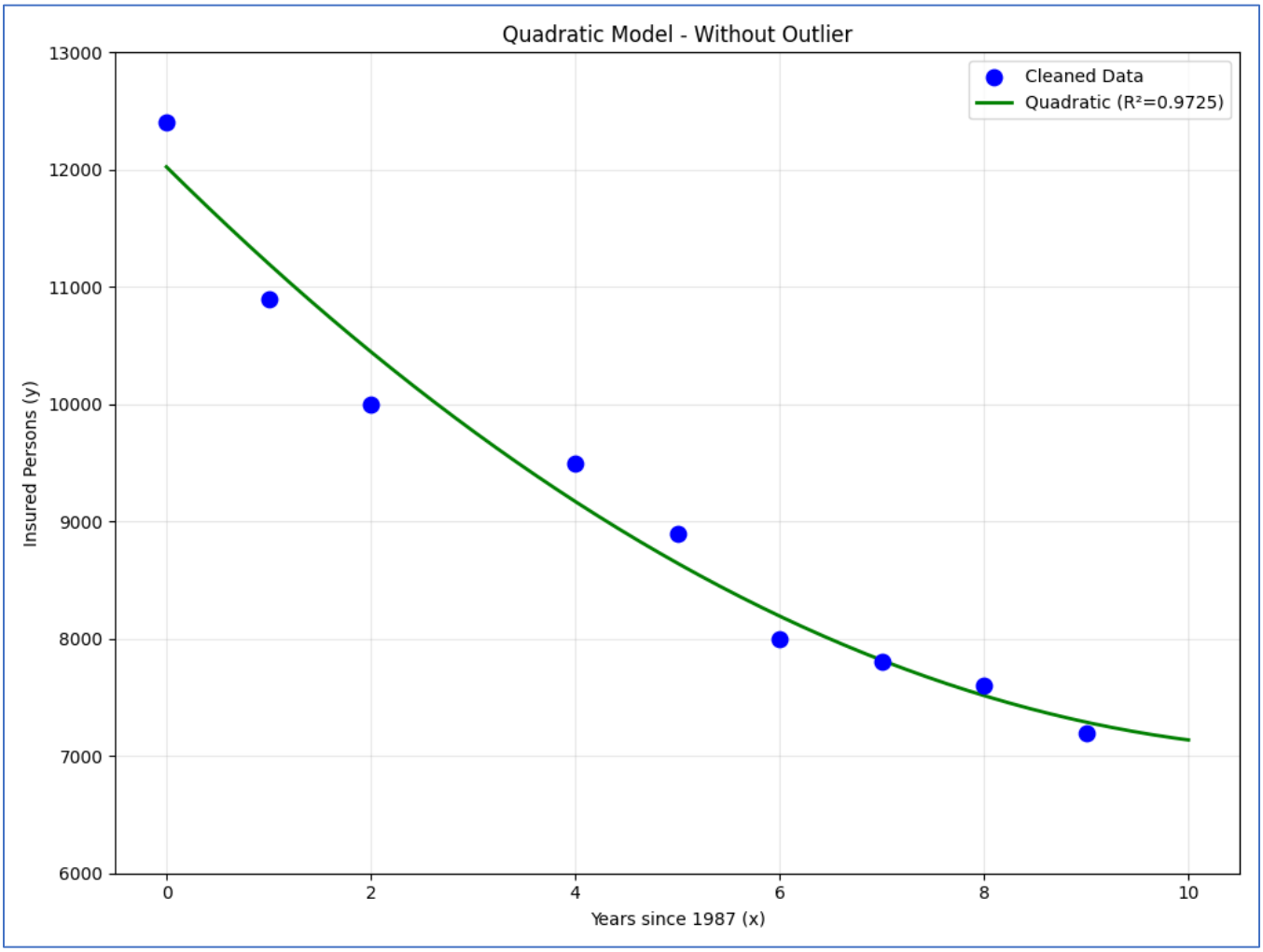

Figure 11: Quadratic Model of Cleaned Data without Outlier

➢ Simulation Results:

```
Quadratic Model:
  Equation: y = 12024.17 + (-863.13)*x^1 + (37.44)*x^2
  R^2: 0.9725
```

Figure 12: Simulation Results of Quadratic Model of Cleaned Data without Outlier

2.3 Cubic Model

The equation for the cubic model is given as follows:

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$$

➢ Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y}$$

$$\boldsymbol{\beta} = \begin{pmatrix} \boldsymbol{\beta}_0 \\ \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \boldsymbol{\beta}_3 \end{pmatrix}, \quad \mathrm{X} = \begin{pmatrix} 1 & 0^2 & 0^3 \\ 1 & 1^2 & 1^3 \\ 1 & 2^2 & 2^3 \\ 1 & 4^2 & 4^3 \\ 1 & 5^2 & 5^3 \\ 1 & 6^2 & 6^3 \\ 1 & 7^2 & 7^3 \\ 1 & 8^2 & 8^3 \\ 1 & 9^2 & 9^3 \end{pmatrix}, Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \boldsymbol{\beta}_0 \\ \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \boldsymbol{\beta}_3 \end{pmatrix} = \begin{pmatrix} 12{,}204.42 \\ -1{,}221.74 \\ 140.17 \\ -7.46 \end{pmatrix}$$

So ,linear equation is:

$$\hat{y}_i = \mathbf{12{,}204.42} - \mathbf{1{,}221.74}\, x_i + \mathbf{140.17}\, x_i^2 - \mathbf{7.46}\, x_i^3$$

➢ R² Verification Table

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{82{,}300}{9} = 9{,}144.44$$

year	x_i	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$	SSE = $(y_i - \hat{y}_i)^2$	SSyy = $(y_i - \bar{y})^2$
1987	0	12400	12204.4	38,251	10,598,642
1988	1	10900	11115.4	46,392	3,081,975
1989	2	10000	10261.9	68,595	731,975
1991	4	9500	8666.9	174,339	126,420
1992	5	8900	8307.8	54,323	59,753

1993	6	8000	7960.3	94,746	1,309,753
1994	7	7800	7579.7	25,702	1,807,531
1995	8	7600	7121.1	413	2,385,309
1996	9	7200	**82300.0**	6,227	3,780,864
Σ =	**42**	82,300	12204.4	**508,988**	**23,882,222**

$$R^2_{cubic\ model} = \frac{SS - SSE}{SS} = \frac{\mathbf{23,882,222} - \mathbf{508,988}}{\mathbf{23,882,222}} = 0.9787$$

➢ Output Graph

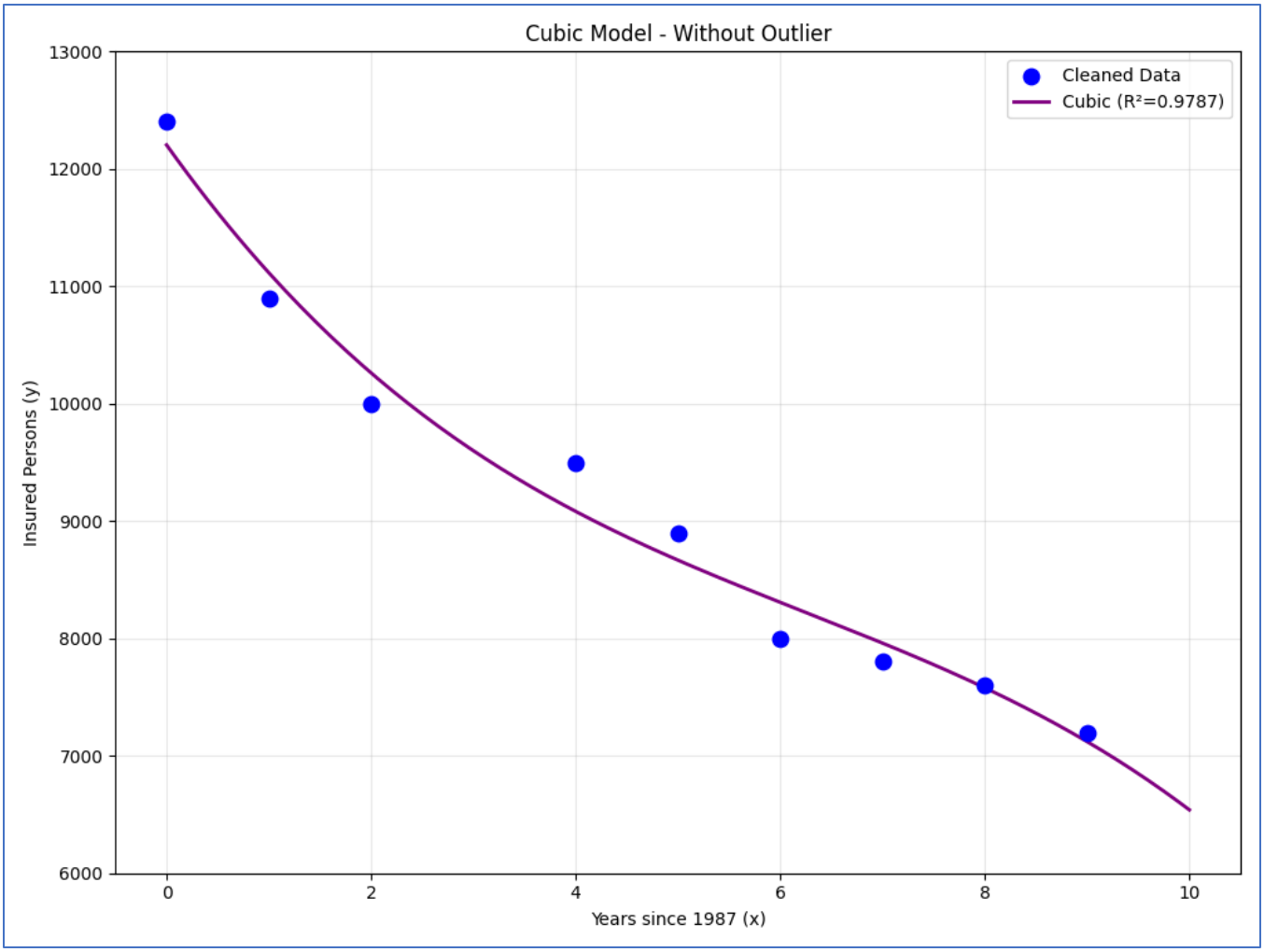

Figure 13:Cubic Model of Cleaned Data without Outlier

➢ Simulation Results:

```
Cubic Model:
  Equation: y = 12204.42 + (-1221.74)*x^1 + (140.17)*x^2 + (-7.46)*x^3
  R^2: 0.9787
```

Figure 14: Simulation Results of Cubic Model of Cleaned Data without Outlier

1.3 Comparison Between Three Models (Cleaned Data):

Model	Equation (MATLAB)	R^2
Linear	$\hat{y}_i = 11{,}621.67 - 530.83\,x$	0.9439
Quadratic	$\hat{y}_i = 12{,}024.17 - 863.13\,x + 37.44\,x^2$	0.9725
Cubic	$\hat{y}_i = 12{,}204.42 - 1{,}221.74\,x_i + 140.17\,x_i^2 - 7.46\,x_i^3$	0.9787

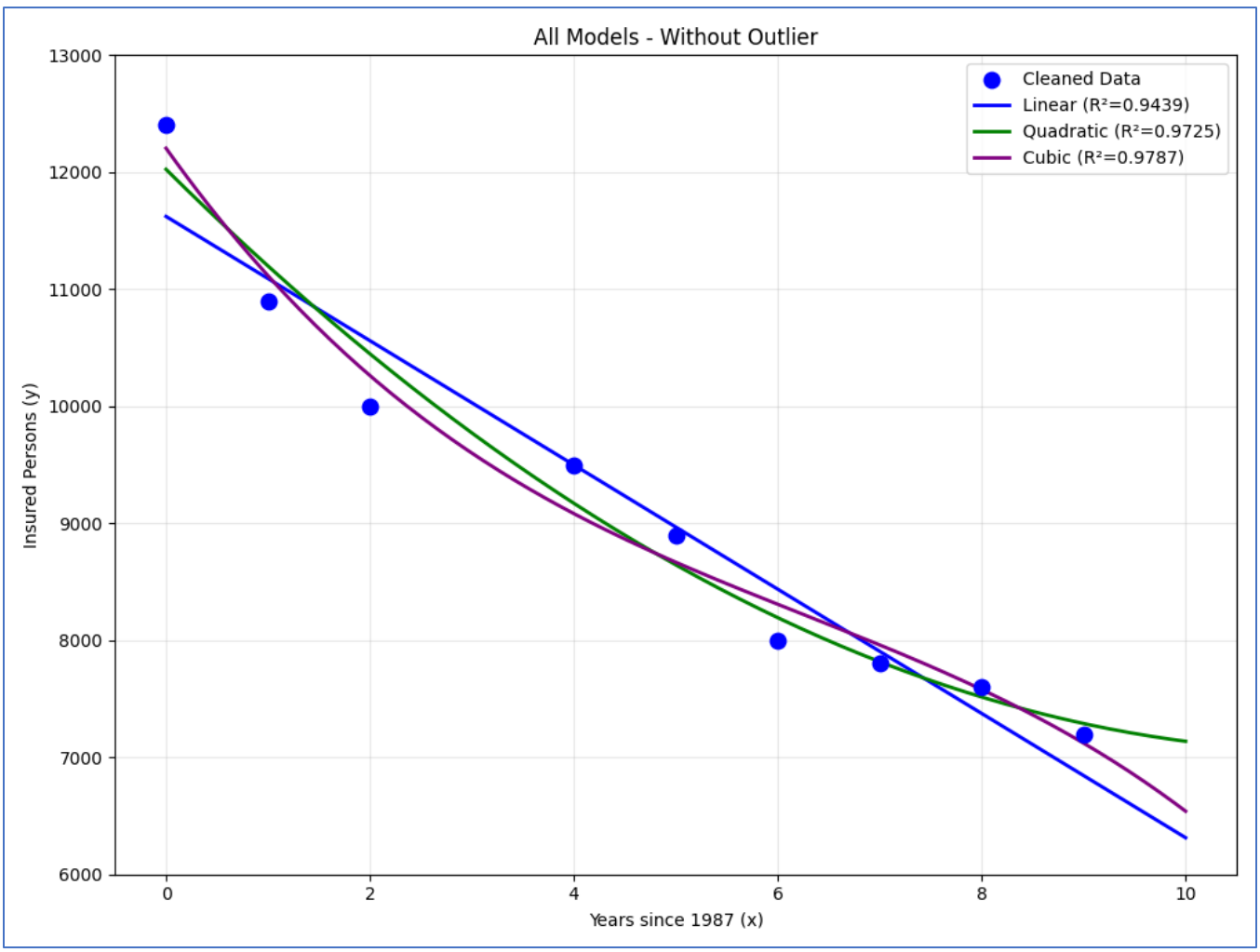

Figure 15: Comparison between three Models of Cleamed Data without Outlier

Comment:

All three R^2 values are very high (> 0.94) after removing the outlier, confirming that it was the primary source of distortion. The linear model alone explains 94.4% of the variance.

1.4 Best-Fit Model Selection

R^2 Comparison

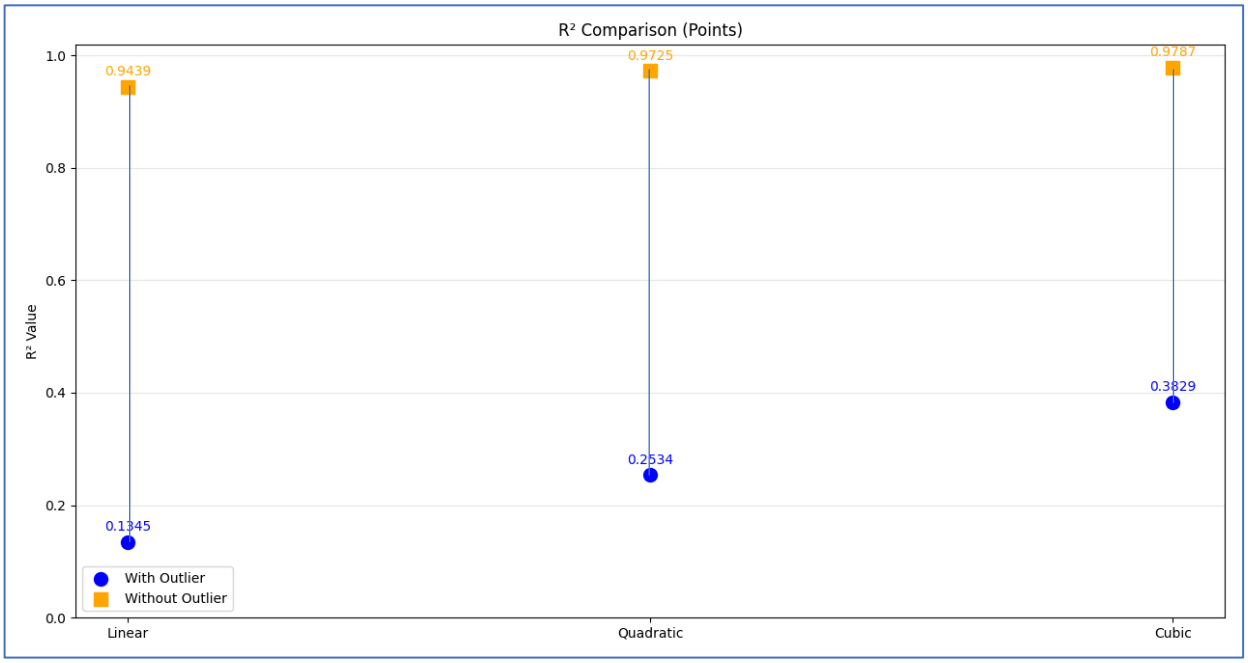

Figure 16: R-square Comparison between three Models

Model	Parameters	R^2 (Original) ⟶ R^2 (Cleaned)	Gain in R^2 (Original) ⟶ Gain in R^2 (Cleaned)
Linear	2	0.1345 ⟶ 0.9439	—
Quadratic	3	0.2534 ⟶ 0.9725	+0.1189 ⟶ +0.0286
Cubic	4	0.3829 ⟶ 0.9787	+0.1295 ⟶ +0.0062

Conclusion (with outlier):

- Linear → Quadratic: R^2 gain = 0.1189 (significant) → Quadratic is better.
- Quadratic → Cubic: R^2 gain = 0.1295 (significant) → Cubic is better.

Best-fit model (with outlier): Cubic $\hat{y} = 13113.15 - 4313.93\,x + 932.31\,x^2 - 58.95\,x^3$ ($R^2 = 0.3829$)

Conclusion (without outlier):

- Linear → Quadratic: R^2 gain = 0.0286 (significant) → Quadratic is better.
- Quadratic → Cubic: R^2 gain = 0.0062 → gain is NOT significant enough.

Best-fit model (without outlier): Quadratic $\hat{y} = 12{,}024.17 - 863.13\,x + 37.44\,x^2$ ($R^2 = 0.972$)

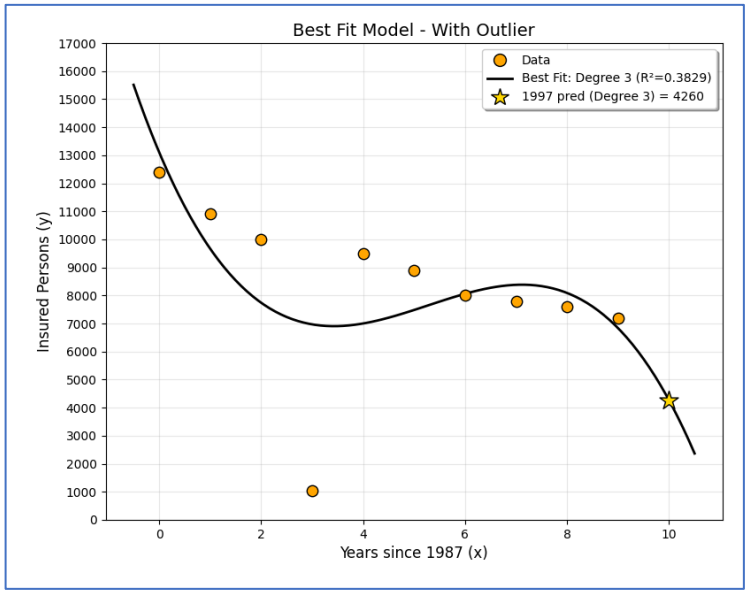

Figure 17: Best Fit Model with Outlier

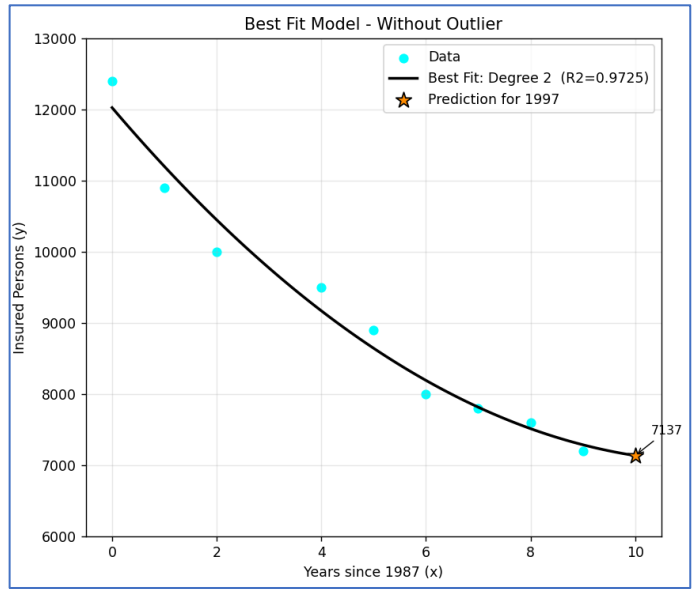

Figure 18: Best Fit Model without Outlier

```
>>  Model Selection: Cubic recommended
   dR2 (Linear -> Quadratic) = +0.1188
   dR2 (Quadratic -> Cubic)  = +0.1295
```

Figure 19: Simulation Results of Best Fit Model with Outlier

```
>>  Model Selection: Quadratic recommended
   dR2 (Linear -> Quadratic) = +0.0286
   dR2 (Quadratic -> Cubic)  = +0.0062
```

Figure 20: SImulation Results of Best Fit Model without Outlier

1.5 Prediction of Insured People

At year 1997 (x=10) :

- Using the best-fit model from the original data with outlier (Cubic model as selected):

$$\hat{y}_i = 13113.15 - 4313.93\,x_i + 932.31\,x_i^2 - 58.95\,x_i^3$$

$\hat{y} = 13113.15 - 4313.93(10) + 932.31(10^2) - 58.95(10^3) = 4{,}254.85$

Insured People ≈ 4,254.85 insured person

- Using the best-fit model from the cleaned data without outlier (Quadratic model as selected):

$$\hat{y}_i = 12{,}024.17 - 863.13\,x + 37.44\,x^2$$

$$\hat{y} = 12{,}024.17 - 863.13(10) + 37.44(10^2) = 7{,}136.87$$

Insured People ≈ 7,136.87 insured person

```
==============================================================
  PREDICTIONS FOR 1997  (year = 1997 , x = 10)
==============================================================
  Original Data with-outlier (Cubic Model) -> 4,260.00 insured persons

  Cleaned Data without-outlier (Quadratic Model) -> 7,137.06 insured persons
```

Figure 21: Simulation Results of Predection Insured People at year 1997

Problem 2: Heating Oil

We develop a regression model to estimate heating oil usage for a single-family home in January, based on two predictors: average temperature (Temp, in °F) and amount of insulation (Insulation, in inches).

2.1 Dataset

Oil (gallons)	Temp (°F)	Insulation (in)
270	40	4
362	27	4
162	40	10
45	73	6
91	65	7
233	**65**	**40 ← OUTLIER**
372	10	6
305	9	10
234	24	10
122	65	4
25	66	10
210	41	6
450	22	4
325	40	4
52	60	10

2.2 Case A: WITH Outlier

1. Scatter Plot & Outlier Identification

A 3D scatter plot of Oil Usage vs. Temperature and Insulation reveals that one data point is far outside the cluster formed by the other observations: the point (Temp=65°F, Insulation=40 in, Oil=233 gallons). All other insulation values range from 4 to 10 inches; an insulation thickness of 40 inches is physically unreasonable for a typical single-family home and is almost certainly a data entry error (e.g., the value 4 recorded as 40). This point is identified as the outlier and is highlighted in red in the scatter plot below.

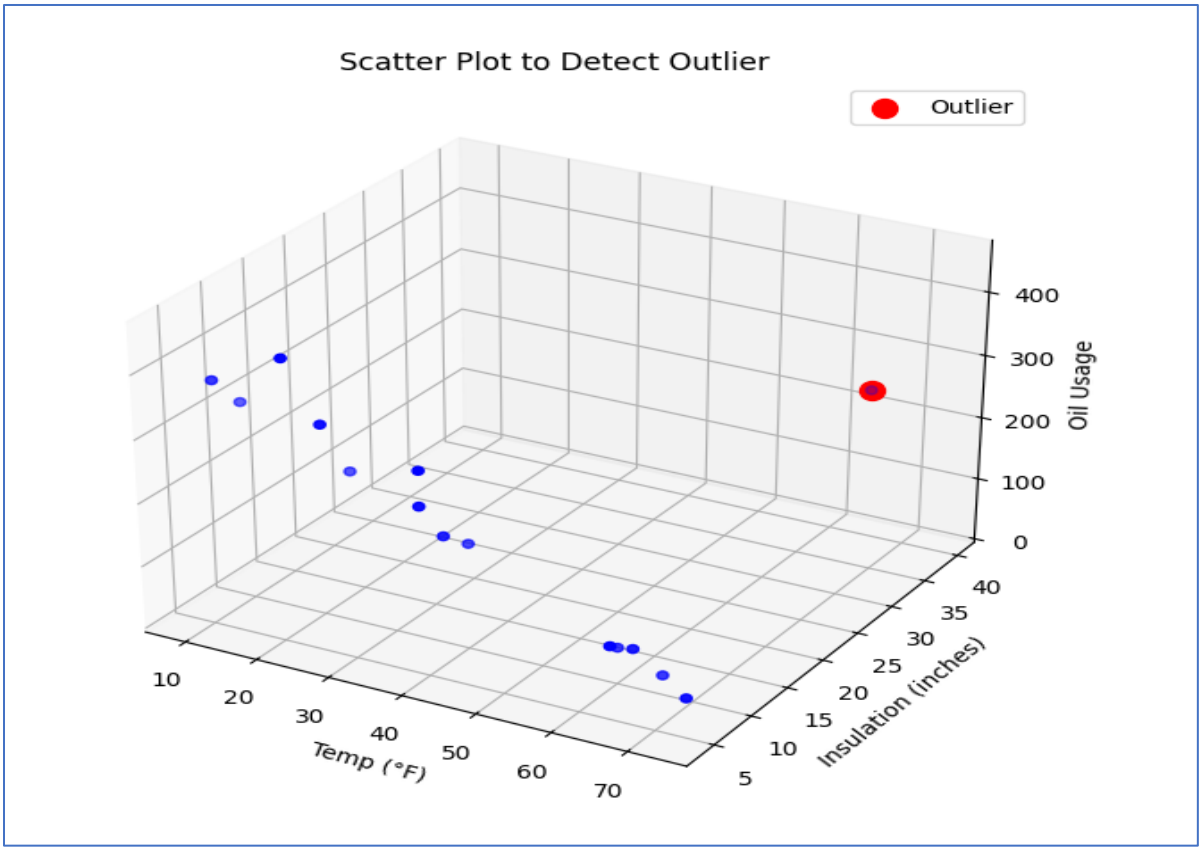

Figure 22: 3D Scatter Plot to Detect Outlier (outlier in red: Insulation=40)

2. Regression Models

2.1 Linear Model

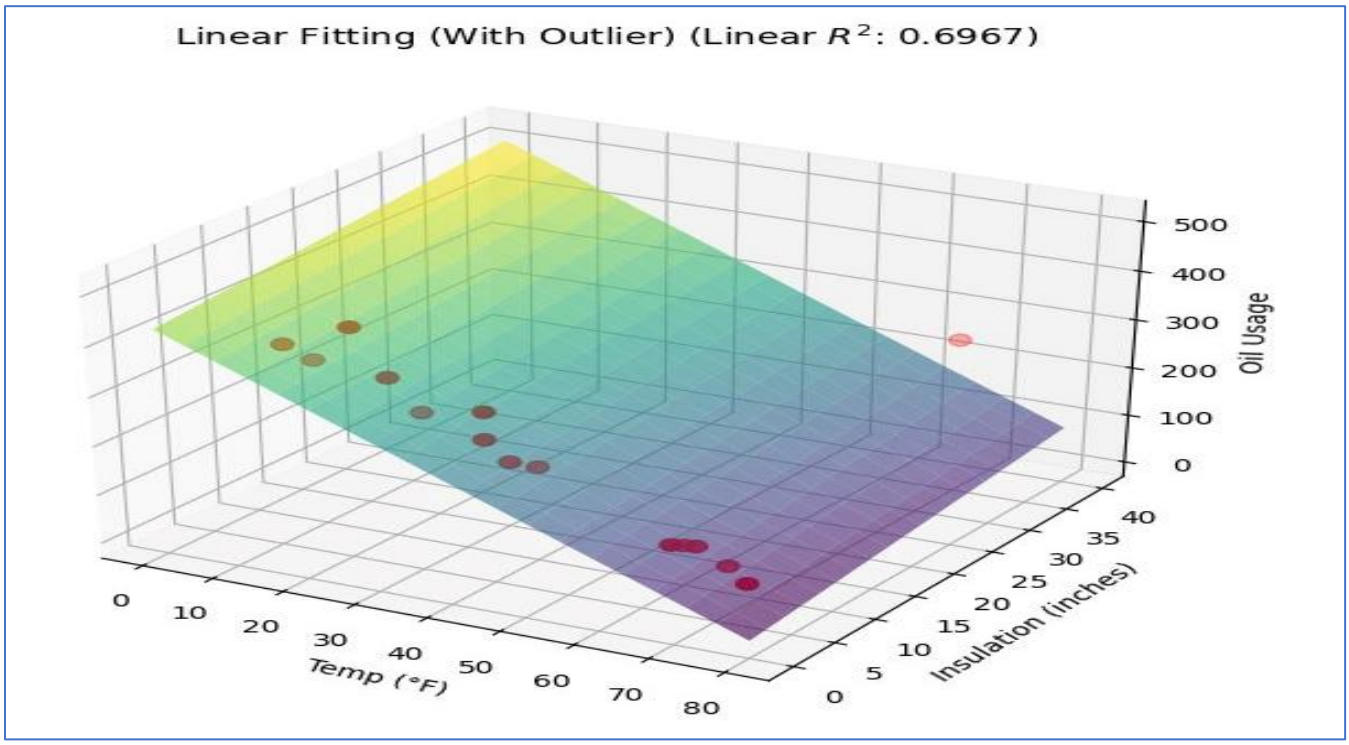

Figure 23: 3D Linear Regression Surface (With Outlier), R² = 0.6967

The multivariate linear model has the form:

$$Oil = \beta_0 + \beta_1 * T + \beta_2 * I$$

where T = Temp, I = Insulation

$$Oil = 426.97 - 5.274 * T + 1.966 * I$$

```
Linear model:
  Intercept : 426.967656
  Coef      : Temp=-5.273465,  Insulation=1.966053
  Equation  : y = 426.9677  +  (-5.2735)*Temp  +  (1.9661)*Insulation
  R2        : 0.696686
```

Figure 24: Simulation Results of Linear Regression Surface (With Outlier)

So, the linear fit equation for the original dataset is:

$$\hat{y}_i = 426.97 - 5.27x_{1i} + 1.966\ x_{2i}$$

The Coefficient of Determination (R²):

R² (Linear, With Outlier) = 0.6967

2.2 Quadratic Model

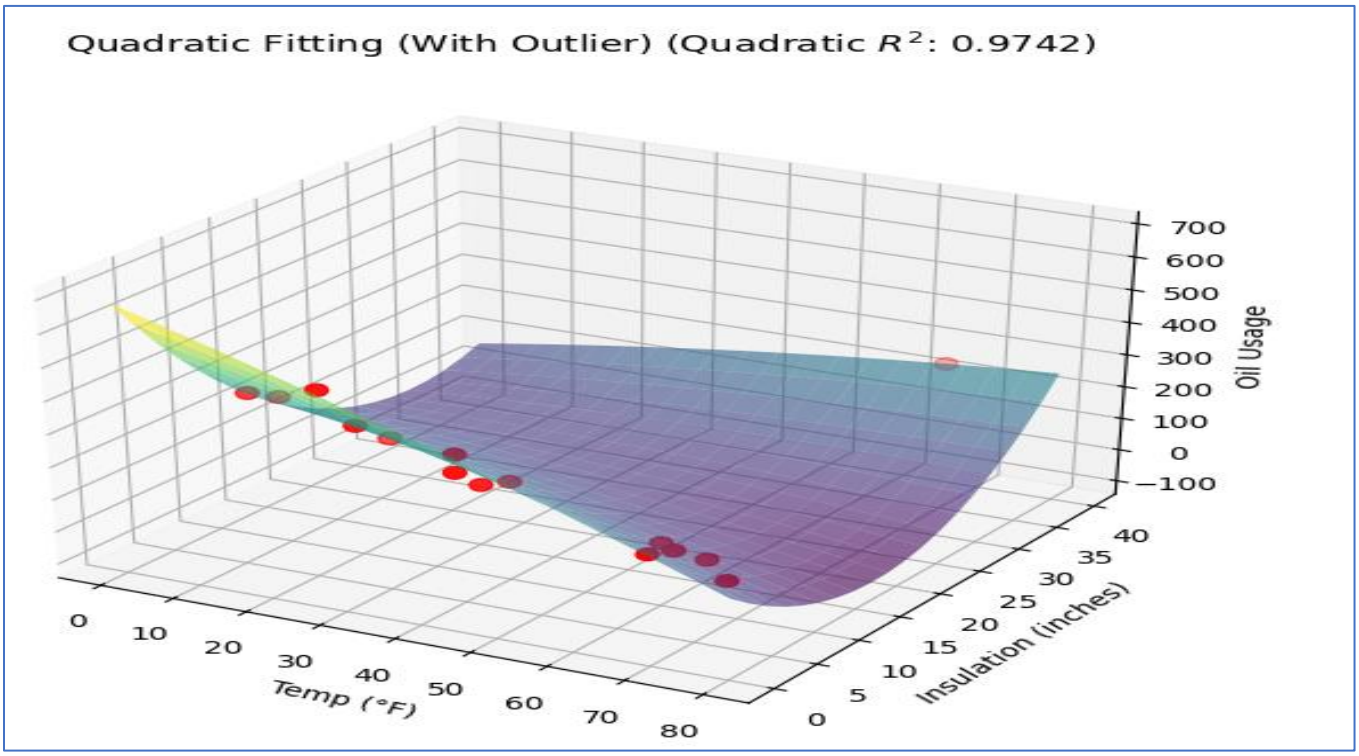

Figure 25: 3D Quadratic Regression Surface (With Outlier), R² = 0.9742

The multivariate linear model has the form:

$$Oil = \beta_0 + \beta_1 T + \beta_2 I + \beta_3 T^2 + \beta_4 TI + \beta_5 I^2$$

where T = Temp, I = Insulation

$$Oil = 680.03 - 6.168\,T - 41.716\,I - 0.01248\,T^2 + 0.2392\,TI + 0.6586\,I^2$$

```
Quadratic model:
  Intercept : 680.025958
  Temp                : -6.167784
  Insulation          : -41.716060
  Temp^2              : -0.012475
  Temp Insulation     : 0.239147
  Insulation^2        : 0.658593
  Equation  : y = 680.0260  +  (-6.1678)*Temp  +  (-41.7161)*Insulation  +  (-0.0125)*Temp^2  +  (0.2391)*Temp
  Insulation  +  (0.6586)*Insulation^2
  R2        : 0.974152
```

Figure 26: Simulation Results of Quadratic Regression Surface (With Outlier)

So, the quadratic fit equation for the original dataset is:

$$\hat{y}_i = 680 - 6.17x_{1i} - 41.72\,x_{2i} - 0.012\,x_{1i}^2 + 0.24\,x_{1i}x_{2i} + 0.66\,x_{2i}^2$$

The Coefficient of Determination (R²):

R² (Quadratic, With Outlier) = 0.9742

2.3 Case B: WITHOUT Outlier

1. Scatter Plot & Outlier Identification (WITHOUT Outlier)

The data point (Temp=65, Insulation=40, Oil=233) is removed. All remaining 14 data points have insulation values between 4 and 10 inches, which is physically reasonable. The cleaned scatter plot is shown below.

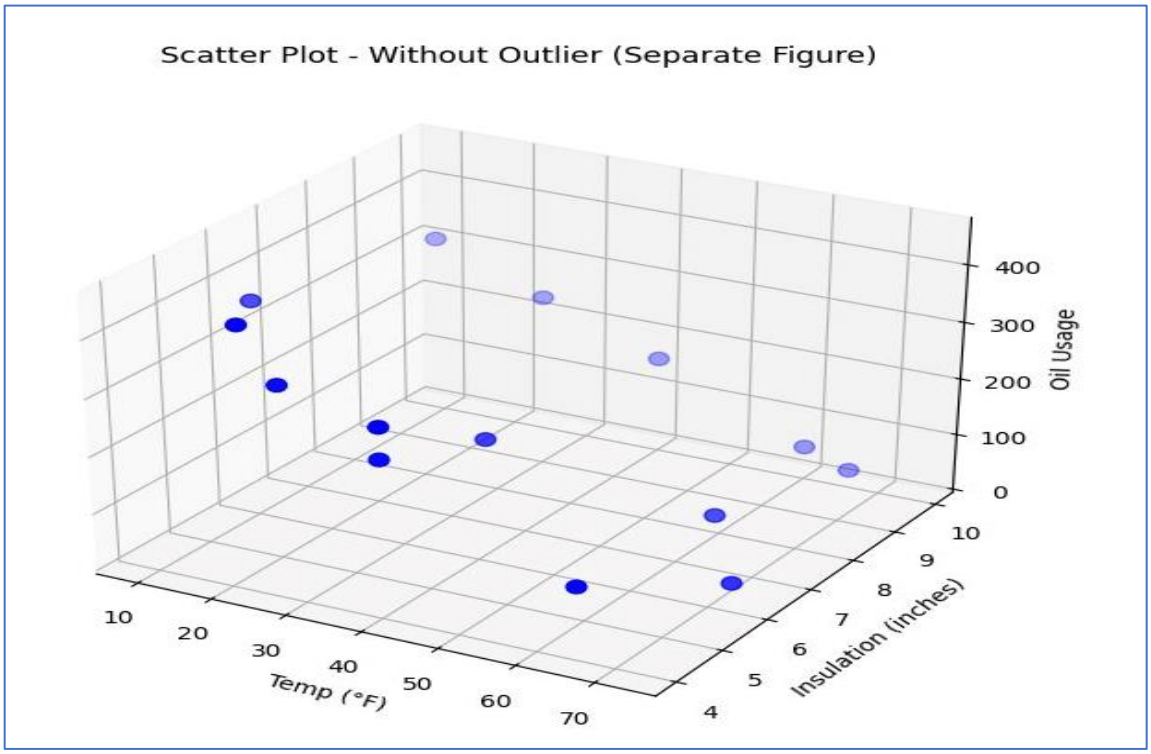

Figure 27: 3D Scatter Plot (Cleaned Data, Outlier Removed)

2. Regression Models (WITHOUT Outlier)

2.1 Linear Model

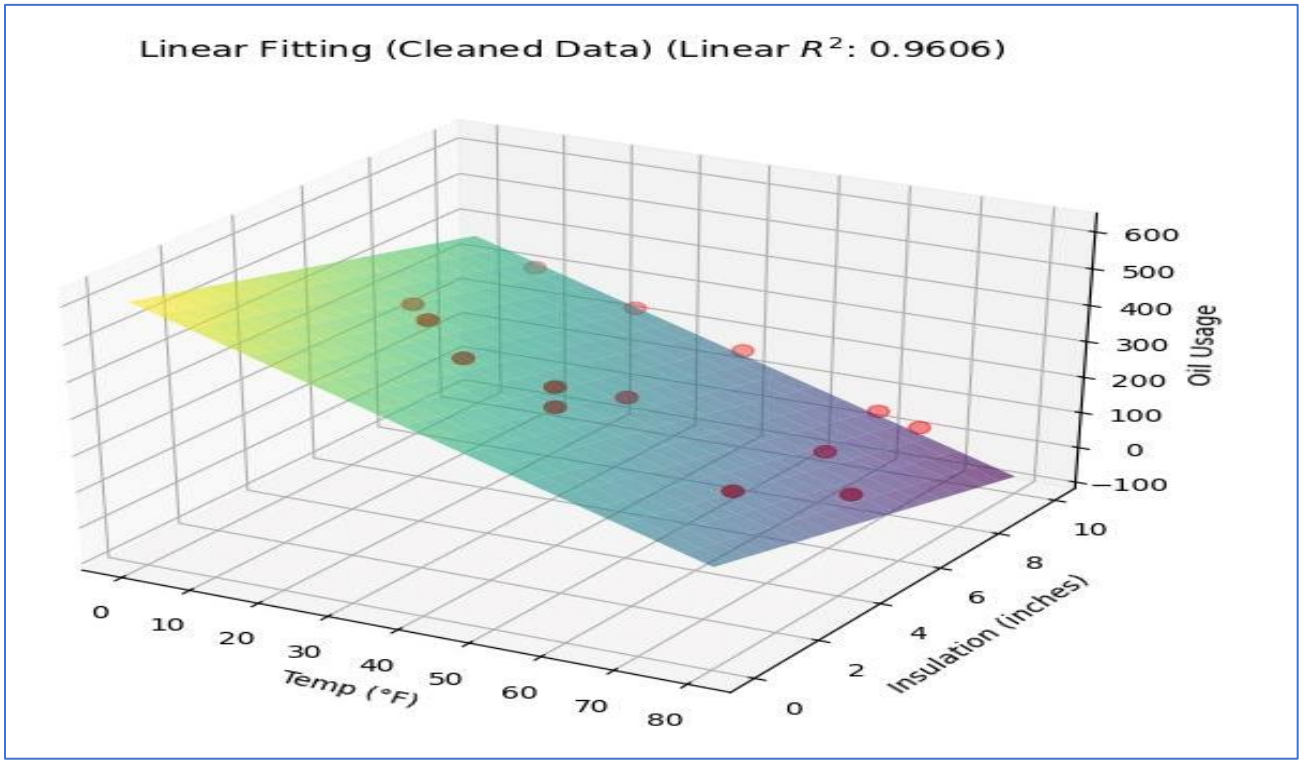

Figure 28: 3D Linear Regression Surface (Cleaned Data), R² = 0.9606

The multivariate linear model has the form:

$$Oil = \beta_0 + \beta_1 * T + \beta_2 * I$$

where T = Temp, I = Insulation

$$Oil = 601.6 - 5.482\,T - 23.232\,I$$

```
Linear model:
  Intercept : 601.601412
  Coef      : Temp=-5.481747,  Insulation=-23.232031
  Equation  : y = 601.6014  +  (-5.4817)*Temp  +  (-23.2320)*Insulation
  R2        : 0.960609
```

Figure 29: Simulation Results of Linear Regression Surface (Cleaned Data)

So, the linear fit equation for the original dataset is:

$$\hat{y}_i = 601.6 - 5.482\,x_{1i} + 23.232\,x_{2i}$$

The Coefficient of Determination (R²):

R² (Linear, With Outlier) = 0.96061

2.2 Quadratic Model

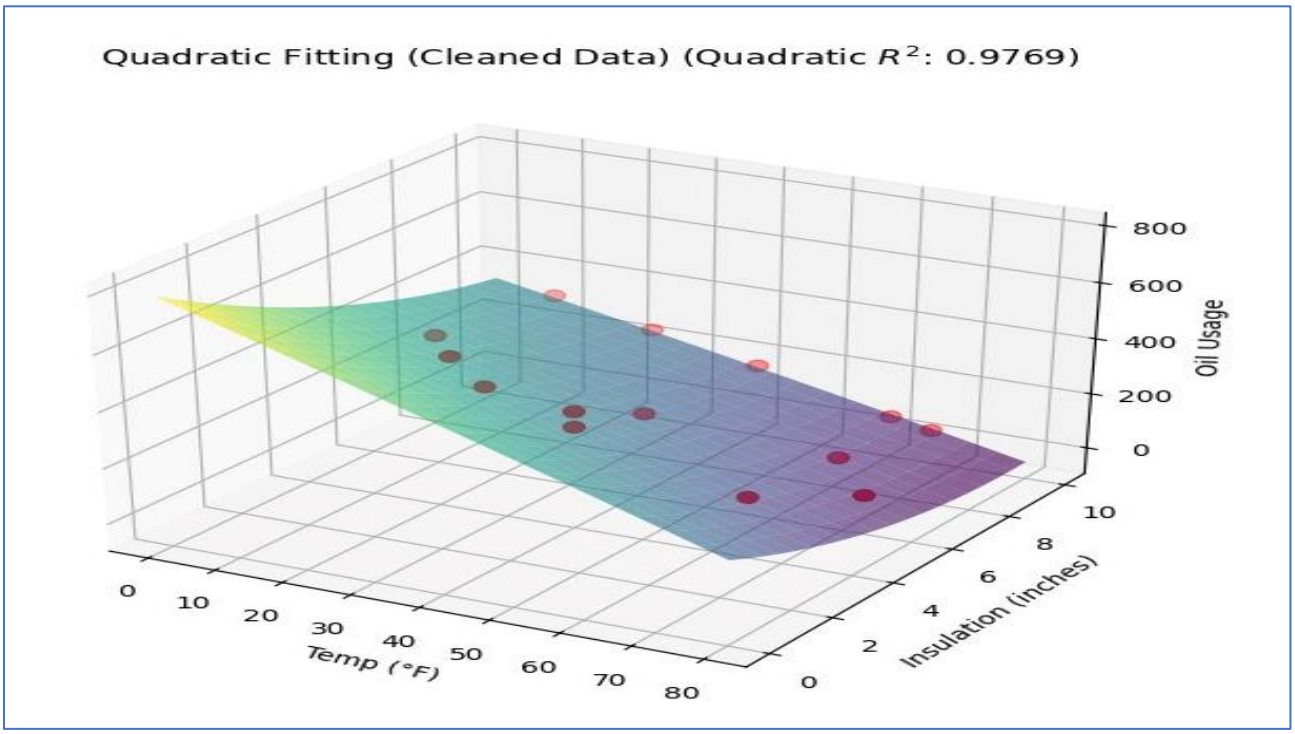

Figure 30: 3D Quadratic Regression Surface (Cleaned Data), R² = 0.9769

The multivariate quadratic model has the form:

$$Oil = \beta_0 + \beta_1 T + \beta_2 I + \beta_3 T^2 + \beta_4 TI + \beta_5 I^2$$

where T = Temp, I = Insulation

$$Oil = 785.52 - 6.963\,T - 72.159\,I - 0.00348\,T^2 + 0.2567\,TI + 2.7438\,I^2$$

```
Quadratic model:
  Intercept : 785.516872
  Temp                : -6.962462
  Insulation          : -72.158742
  Temp^2              : -0.003478
  Temp Insulation     : 0.256701
  Insulation^2        : 2.743800
  Equation  : y = 785.5169  +  (-6.9625)*Temp  +  (-72.1587)*Insulation  +  (-0.0035)*Temp^2  +  (0.2567)*Temp
  Insulation  +  (2.7438)*Insulation^2
  R2        : 0.976889
```

Figure 31: Simulation Results of Quadratic Regression Surface (Cleaned Data)

So, the linear fit equation for the original dataset is:

$$\hat{y}_i = 785.52 - 6.963\,x_{1i} - 72.159\,x_{2i} - 0.00348\,x_{1i}^2 + 0.2567\,x_{1i}x_{2i} + 2.7438\,x_{2i}^2$$

The Coefficient of Determination (R²):

R² (Quadratic, With Outlier) = 0.9769

2.4 Comparison of Models and Best-Fit Selection

Model	R^2 (With Outlier)	R^2 ² (Without Outlier)	Gain (With Outlier)	Gain ² (Without Outlier)
Linear	0.6967	0.9606	—	—
Quadratic	0.9742	0.9769	+0.2775	+0.0163

Conclusion (Part a — With Outlier):
Linear → Quadratic gain = +0.2775

Best-fit model (with outlier): Quadratic (R^2 = 0.9742).

Conclusion (Part b — With Outlier):
Linear → Quadratic gain = +0.0163 (NOT significant enough).

Best-fit model (without outlier): Linear (R^2 = 0.9606).

After removing the outlier, the linear model already achieves R^2 = 0.9606. The gain from linear to quadratic is only +0.0163, which is NOT significant enough as specified in Note 4 of the assignment.

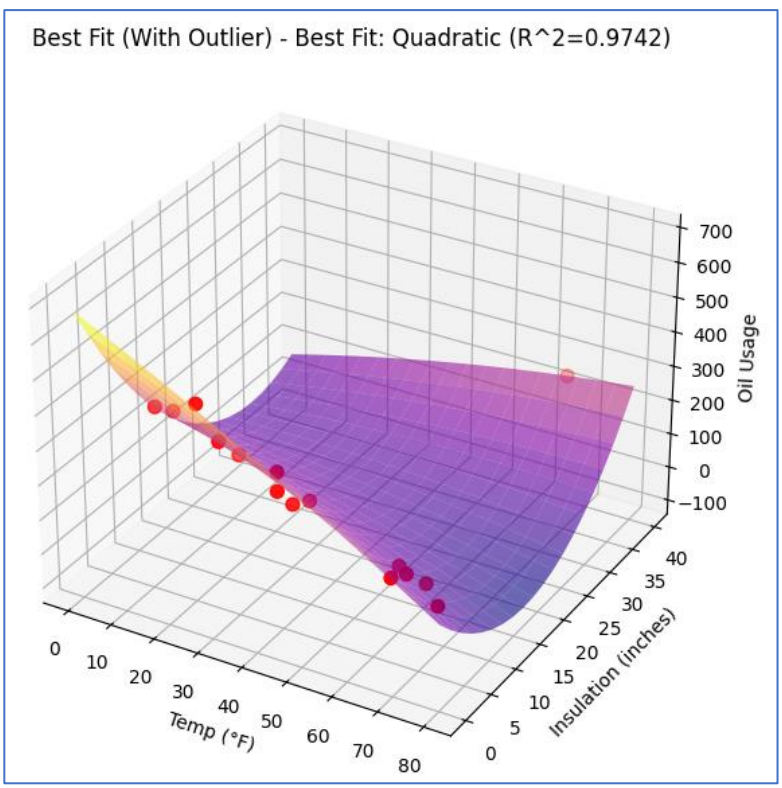

Figure 33: Best Fit Model of Original Data (Quadratic Model)

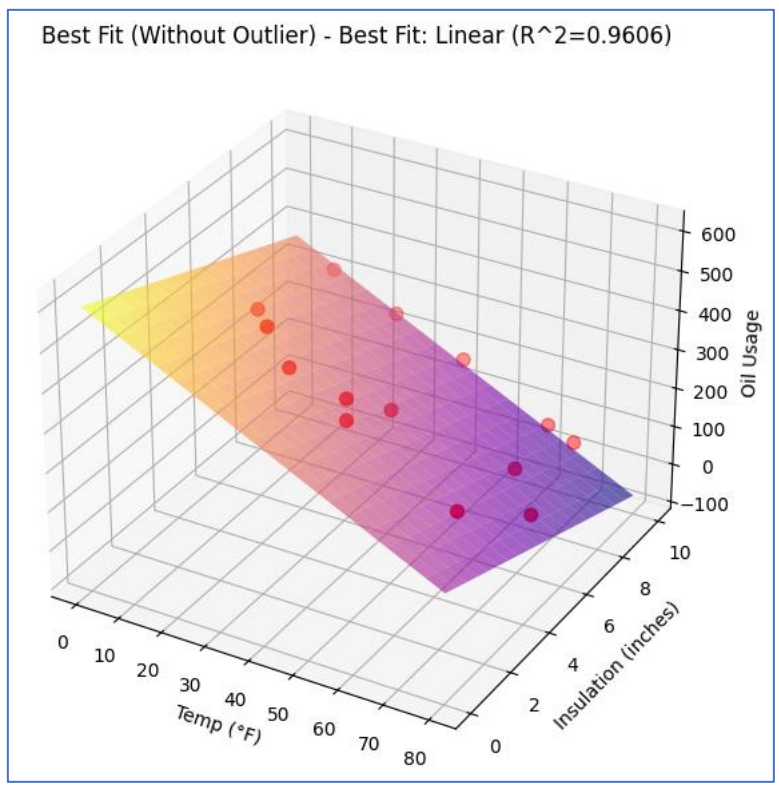

Figure 32: Best Fit Model of Cleaned Data (Linear Model)

➢ **Simulation Results:**

```
Part a  (With Outlier):
  dR2 (Linear -> Quadratic) = +0.2775  -->  significant gain
  Best-fit model: Quadratic  (R2 = 0.9742)

Part b  (Without Outlier):
  dR2 (Linear -> Quadratic) = +0.0163  -->  not significant
  Best-fit model: Linear     (R2 = 0.9606)
```

Figure 34: Simulation Results of Best Fit Model

2.5 Prediction of Oil

Temp =15°F, Insulation = 5 inches

The linear model could be abstracted as follows:

$$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

Linear Model	Data with Outlier	Data without Outlier (Best Fit)
β_0	427	601.6
β_1	- 5.273	- 5.482
β_2	1.966	- 23.23
R^2	0.6966	0.9606
$\hat{y}$(temp = 15, insulation = 5)	357.7	403.22

- Using the best-fit model from the cleaned data without outlier (Linear model as selected):

Oil = 601.60 − 5.482 * 15 − 23.232 * 5 = 403.21

Predicted Oil Usage ≈ 403.2 gallons

➢ **Simulation Results:**

```
Prediction at (15F,5in) using CLEANED Linear best-fit: 403.22 units
```

Figure 35: Simulation Results of Prediction Oil of Cleaned Data

The quadratic model could be abstracted as follows:

$$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1}^2{}_i + \beta_4 x_{1i} x_{2i} + \beta_5 x_{2}^2{}_i$$

Linear Model	Data with Outlier (Best Fit)	Data without Outlier
β_0	680	785.5
β_1	- 6.17	- 6.96
β_2	- 41.72	- 72.16
β_3	- 0.012	- 0.0034
β_4	0.24	0.2567
β_5	0.66	2.744
R^2	0.9741	0.9768
$\hat{y}$(temp = 15, insulation = 5)	410.52	407.35

- Using the best-fit model from the original data with outlier (Quadratic model as selected):
  Oil = 680 + (−6.17)(15) + (−41.72)(5) + (−0.012)(15²) + (0.24)(15)(5) + (0.66)(5²) = 410.65

Predicted Oil Usage ≈ 410.65 gallons

➢ **Simulation Results:**

```
Prediction at (15F,5in) using ORIGINAL Quadratic best-fit: 410.52 units
```

Figure 36: Simulation Results of Prediction Oil of Original Data